Hotel Review Sentiment & Aspect Rating Analysis

MACHINE LEARNING • MINI PROJECT

Predicting overall and aspect-specific hotel ratings from TripAdvisor review text

JIGNAS , PES2UG24AM110
PRASANNA, PES2UG24AM143

OBJECTIVE

Problem: infer a hotel guest's rating from the language used in a review, and examine whether the same text can support aspect-level predictions.

- Predict the overall 1–5 star rating from review text.
- Convert star ratings into negative / neutral / positive polarity.
- Study individual aspects: rooms, location, value, cleanliness, and sleep quality.
- Compare text representations, algorithms, and class-balancing strategies.

DATASET & PREPROCESSING

1,000
usable hotel reviews

5
aspect labels

Source: public TripAdvisor hotel-review sample used as a smaller substitute for the original study dataset.

Text: review title + review body combined into one input field.

Labels: overall rating plus rooms, location, value, cleanliness and sleep quality.

Cleaning: numeric 1–5 ratings validated; empty text and missing overall ratings removed.

Split: random 2/3 training and 1/3 test, using a fixed random seed.

CLASS IMBALANCE

Strategy	Star acc.
No balance	55.4%
Class weight	57.2%
Oversampling	55.7%
undersampling	47.9%

END-TO-END METHOD



FEATURE ENGINEERING

Word unigrams: individual words as features.

Character 5-grams: short character sequences that can capture word fragments and spelling patterns.

Word 2/3-grams: short phrases that preserve local context.

TF-IDF: down-weights very common words and emphasizes more informative terms.

Stop-word removal: applied to word-based representations.

MODELS

Multinomial Naive Bayes: probabilistic baseline for text counts.

Linear SVM: discriminative classifier for high-dimensional sparse text features.

Linear Regression: predicts a continuous star score; predictions are rounded/clipped to 1–5.

The rating distribution is skewed, so four training strategies were compared using word-unigram TF-IDF + Linear SVM.

Finding: class weighting was the strongest balancing strategy in this experiment, improving exact-star accuracy to 57.2%. Undersampling discarded too much information.

EVALUATION

Star accuracy: exact 1–5 rating match.

Polarity accuracy: 1–2 = negative, 3 = neutral, 4–5 = positive.

Balanced accuracy: accounts for performance across imbalanced classes.

Macro F1: gives equal importance to each class.

PROJECT SCOPE NOTE

Important: this is a paper-style reproduction/approximation on a 1,000-review public sample, not an exact reproduction of the original study's 235,793-review dataset. The substitute dataset has five aspect labels and does not contain the original study's full aspect set.

Experimental Results & Findings

Model comparison • feature comparison • aspect-specific analysis • demonstration

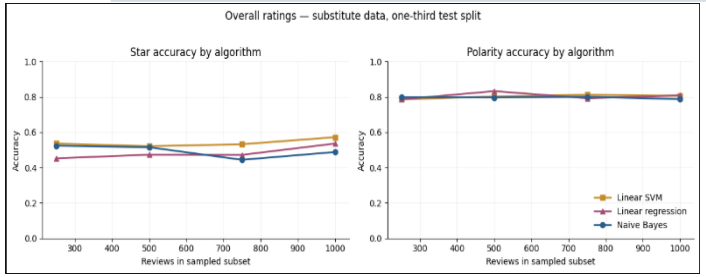
57.2% best exact-star accuracy

81.1% best polarity accuracy

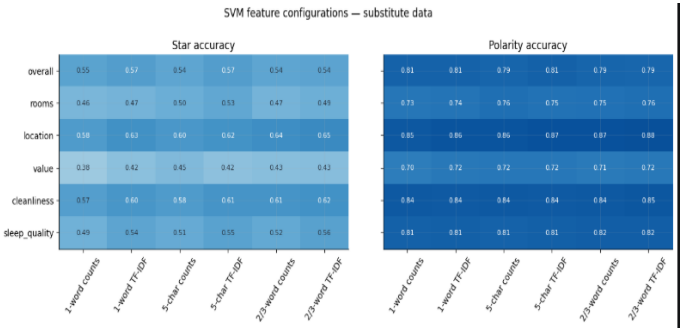
2 / 5 aspects improved by aspect training

SVM strongest exact-star model

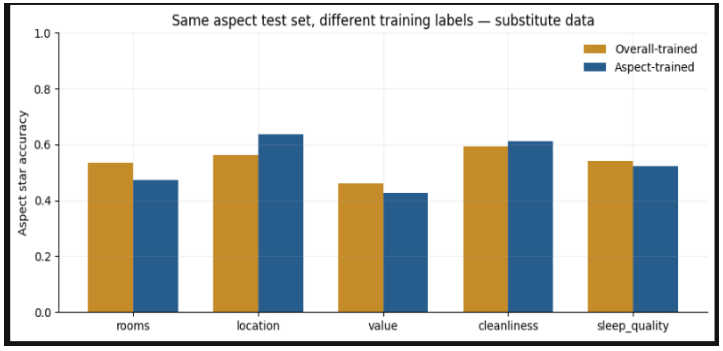
ALGORITHM COMPARISON



FEATURE REPRESENTATION



ASPECT-SPECIFIC MODELLING



KEY RESULTS

Linear SVM: 57.2% exact-star accuracy and 80.5% polarity accuracy.

Naive Bayes: 48.8% exact-star accuracy; useful as a simple baseline.

Linear Regression: 53.6% exact-star accuracy and 80.8% polarity accuracy.

Best feature tie: word-unigram TF-IDF and character-5-gram TF-IDF both reached 57.2% exact-star accuracy.

ASPECT FINDINGS

Location: aspect-trained model improved from 56.3% to 63.8%.

Cleanliness: improved from 59.3% to 61.1%.

Rooms: overall-trained model performed better (53.6% vs 47.3%).

Value: overall-trained model performed better(46.1%vs 42.6%)

Sleep quality: overall-trained model performed better (54.2% vs 52.1%).

LIMITATIONS & FUTURE WORK

Small sample: 1,000 reviews versus the original study's much larger dataset.

Dataset mismatch: the substitute data lacks some original aspect labels.

Random review split: the same hotel may appear in both train and test sets.

Aspect prediction: models can predict an aspect even when the review barely discusses it.

Next: expand the dataset, use hotel-aware splits, add richer error analysis, and compare with transformer-based models.

LIVE DEMONSTRATION

Example input: "Convenient stay"

Target	Predicted	Polarity
Overall	4	Positive
Rooms	5	Positive
Location	5	Positive
Value	3	Neutral
Cleanliness	4	Positive
Sleep- Quality	5	Positive

The demo illustrates the end-to-end workflow: enter review text → vectorize → apply trained model → return overall/aspect rating predictions.

Conclusion

The project demonstrates a complete traditional NLP + machine-learning pipeline for prediction. Text is converted into sparse numerical features , models learn the relationship between language and ratings, and predictions are evaluated at both star and polarity levels.

Main takeaway: on this small sample, class- weighted linear SVM provided the strong exact -star performance, while aspect-specific training was most useful for location and cleanliness

Implementation stack: Python • pandas • NumPy • scikit-learn • Matplotlib